TA Practice Sheet

10 October 2026 16:32

Statistics for ML

TA Practice Sheet

Level: Introductory Undergraduate · Focus: LLN & CLT (Sums & Means)

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Q2

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Q2. Sample Mean and LLN

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with finite mean μ. Define $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. According to the Law of Large Numbers, what happens to $\bar{X}_n$ as $n \to \infty$?

Given $X_1, X_2, \ldots, X_n$ iid with finite mean $E(X_i) = \mu$

iid → independently & identically distributed

Sample mean $= \bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$

By LLN, $S_n \xrightarrow{P} \mu$

- From the Law of Large Numbers, $\frac{S_n}{n} \longrightarrow \mu$,
  - Intuitively, if we take a large number of samples, the sample mean will 'approach the true mean.

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Q1

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Part A — Law of Large Numbers

Q1. Coin Tosses

A fair coin is tossed independently n times. Let $\hat{p}_n$ be the fraction of tosses that result in heads. According to the Law of Large Numbers, what does $\hat{p}_n$ approach as $n \to \infty$?

Let $X_i = 1$, if toss i heads and $X_i = 0$ otherwise

$X_i \sim$ Bernoulli(0.5) (Assuming coin is unbiased)

$E(X_i) = p = 0.5$

Fraction of heads is $\hat{p_n} = \frac{1}{N} \sum_{i=1}^{N} X_i$

By LLN, $\hat{p_n} \xrightarrow{P} 0.5$ ∴ As no. of tosses increases, fractions of heads approach to 0.5

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Q3

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Part B — CLT for Sums

Q3. A Very Basic Sum

Let $X_1, \ldots, X_{100}$ be i.i.d. random variables with mean 2 and standard deviation 3. Define $S = \sum_{i=1}^{100} X_i$. Find the mean and standard deviation of S.

$S = \sum_{i=1}^{100} X_i$; $E(X_i) = 2$, $Var(X_i) = 3^2$, $\sigma_{X_i} = 3$

$E(S) = \sum_{i=1}^{100} E(X_i) = 100 \times 2 = 200$ //

$Var(S) = \sum_{i=1}^{100} Var(X_i) = 100 \times 3^2 = 900$ //

$\sigma_S = \sqrt{Var S} = 30$ //

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Q4. Sum of Bernoulli Random Variables

Let $X_1, \ldots, X_{100}$ be independent Bernoulli random variables with success probability $p = 0.5$. Define $S = \sum_{i=1}^{100} X_i$. Using the CLT, approximate $P(S > 55)$.

$E(X_i) = p = 0.5 \qquad Var(X_i) = p(1-p) = 0.25$

$E(S) = \sum_{i=1}^{100} E(X_i) = 100 \times 0.5 = 50$

$Var(S) = \sum_{i=1}^{100} Var(X_i) = 100 \times 0.25 = 25$

$N(\mu, \sigma^2) \rightarrow N(0,1)$

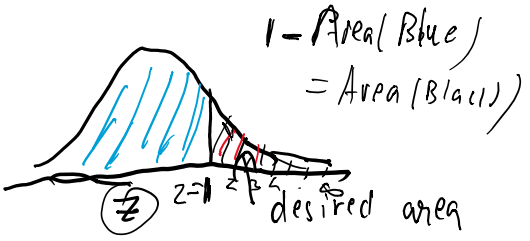
$Z = \frac{X-\mu}{\sigma}$

By CLT, $S \sim N(50, 5^2)$

$P(S > 55) \approx P\left(Z > \frac{55-50}{5}\right)$

$= P(Z > 1) = 1 - \Phi(1)$

where Φ is the cdf of std normal dist^n



$P(Z > 1) = 1 - \Phi(1)$

$\Phi = F_X(x)$

$P(S < 50) = P\left(Z < \frac{50-50}{5}\right)$

$= P(Z < 0) = 1/2$



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Q5

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Q5. Sum of Dice

Ten fair six-sided dice are rolled independently. Use the Central Limit Theorem to approximate the probability that the sum of the values obtained is between 30 and 40, inclusive.

$E(X_i) = \frac{1}{6}\left(\sum_{k=1}^{6} k\right) = 3.5$ // ; X_i (iid)

$\text{Var}(X_i) = 35/12$: $\{\text{Var}(X_i) = E(X_i^2) - (E(X_i))^2\}$

$\frac{1}{6}(1+2+3+..+6)$

$= \frac{1}{6} \times \frac{6 \times 7}{2} = 3.5$

$E(S) = \sum_{i=1}^{10} E(X_i) = 10 \times 3.5 = 35$

$\text{Var}(S) = \sum_{i=1}^{10} \text{Var}(X_i) = 10\left(\frac{35}{12}\right) = \frac{175}{6}$

$\Rightarrow \sigma_S = \sqrt{175/6} \approx 5.4$

X be rv

$\frac{1}{6} \times 1 + \frac{1}{6} \times 2 + \ldots + \frac{1}{6} \times 6$

$= \frac{1}{6} \times (1+2+\ldots+6)$

$E(X) = \sum_x x\, p(x)$

By the CLT $S \sim N(35, 5.4^2)$

$P(30 \le S \le 40) \approx P\left(\frac{30-35}{5.4} \le Z \le \frac{40-35}{5.4}\right)$ ~ 0.926

$\Rightarrow P(30 \le S \le 40) \approx \phi(0.926) - \phi(-0.926)$

$= P(Z \le 0.926) - P(Z \le -0.926)$



$P(30 < S < 40)$

↳ continuous rv ○

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Q6

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Q6. Sum of Rounding Errors

Fifty numbers are rounded to the nearest integer and then summed. Assume individual rounding errors are independent and uniformly distributed between −0.5 and 0.5. Using the CLT, approximate the probability that the resulting sum differs from the exact sum by more than 3.

$|S| > 3 \rightarrow S > 3,\ S < -3$

$\sim U(a,b)$ $E(X) = \frac{a+b}{2}$ $Var(X) = \frac{(b-a)^2}{12}$

Let X_i be the rounding error ; $S = \sum_{i=1}^{50} X_i$

$X_i \sim U(-0.5, 0.5) \Rightarrow E(X_i) = \frac{-0.5+0.5}{2} = 0$ //

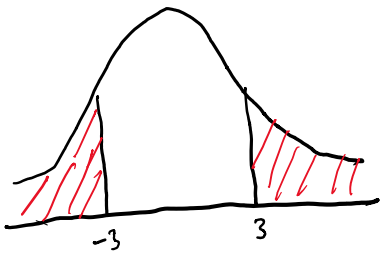
$Var\ X_i = \frac{(0.5-(-0.5))^2}{12} = 1/12$ //

$E(S) = \sum_{i=1}^{50} E(X_i) = 0$

$Var(S) = 50 \times \frac{1}{12} = 25/6$

$P(|S|>3) = P(S<-3) + P(S>3)$

$S \sim N(0, 25/6)$



By symmetry of gaussian dist^n

$P(|S|>3) = 2P(S>3)$

$\approx 2P\left(Z > \frac{3-0}{\sqrt{25/6}}\right)$

$\approx 2[1 - \Phi(1.47)]$

$X_i \sim N(\mu, \sigma)$

$S = \sum_{i=1}^{n} X_i$

$E(S) = \sum_{i=1}^{n} E(X_i) = n\mu$

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$$E(S) = \sum_{i=1}^{n} E(x_i) = n\mu$$

$$\text{Var}(S) = \sum_{i=1}^{n} \text{Var}(x_i) = n\sigma^2$$

$$S = \alpha \qquad Z > \frac{\alpha - n\mu}{\sqrt{n\sigma^2}} \Rightarrow Z > \frac{\alpha - n\mu}{\sigma\sqrt{n}}$$

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Q7

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Q7. Insurance Claims

An insurance company has 25,000 automobile policyholders. Yearly claims have mean 320 and standard deviation 540 (i.i.d.). Use the CLT to approximate the probability that total yearly claims exceed 8.3 million.

$$S = \sum_{i=1}^{n} X_i$$

$$E(S) = \sum_{i=1}^{25,000} E(X_i) = 25000\,(320) = 8,000,000 = 8 \text{ million}$$

$$\text{Var } S = \sum_{i=1}^{25,000} E(X_i) = 25000 \times (540)^2$$

$$\Rightarrow \sigma_S = \sqrt{\text{Var } S} = 540\sqrt{25000} = 85,381$$

$$P(S > 8,300,000) = P\left(Z > \frac{8,300,000 - 8,000,000}{85,381}\right)$$

$$= P(Z > 3.51) = 1 - \Phi(3.51)$$

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Q8

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Part C — CLT for Sample Means

Q8. A Very Basic Sample Mean

Let $X_1, \ldots, X_{25}$ be i.i.d. random variables with mean 10 and standard deviation 5. Define $\bar{X} = \frac{1}{25}\sum_{i=1}^{25} X_i$. Find the mean and standard deviation of $\bar{X}$.

$\sigma_x = \sqrt{\text{Var}\,x}$; $\sigma_x^2 = \text{Var}\,x$ $\quad \text{Var}(aX+b) = a^2\text{Var}(X)$ + Var b → 0

$x_1, x_2, \ldots, x_n$ → identically distributed → same pop (i.i.d)

$$E(\bar{x}) = E\left(\frac{1}{25}\sum_{i=1}^{25} x_i\right) = \frac{1}{25}E\left(\sum_{i=1}^{25} x_i\right) = \frac{1}{25} \times 25 \times E(x_i)$$

$\Rightarrow E(\bar{x}) = 10$

$\text{Var}(\sum x_i) = \sum \text{Var}\,x$

$$\text{Var}(\bar{x}) = \text{Var}\left(\frac{1}{25}\sum_{i=1}^{25} x_i\right) = \frac{1}{25^2}\sum_{i=1}^{25}\text{Var}(x_i) = \frac{1}{25^2} \times 25 \times \text{Var}(x_i)$$

(n, σ^2)

$$\Rightarrow \text{Var}(\bar{x}) = \frac{1}{25^2} \times 25 \times \sigma^2_{x_i} = \frac{1}{25^2} \times 25 \times 25 = 1$$

$$\Rightarrow \sigma_{\bar{x}} = \sqrt{\text{Var}(\bar{x})} = 1$$

$\sigma^2(1+1+\ldots+1)$? $\quad \frac{n\sigma^2}{n^2} = \sigma^2/n$

Generally for any n $\quad E(x_i) = \mu$, $\text{Var}(x_i) = \sigma^2$

$$E(\bar{x}) = \frac{1}{n}E\left(\sum_{i=1}^{n} x_i\right) = \frac{n\mu}{n} = \mu$$

$$\text{Var}(\bar{x}) = \sum_{i=1}^{n}\text{Var}\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}\,x_i = \frac{1}{n^2}(\sigma^2 \times n) = \sigma^2/n$$

$\frac{1}{n^2}\sum_{i=1}^{n}\text{Var}\,x_i$ $\quad \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 =$

$$\Rightarrow \sigma_{\bar{x}} = \sigma/\sqrt{n} \quad \Rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$$

n → no. of sample

X_i →

$\bar{X}$ → mean of sample

x_i → height of wt

$\bar{X} = \left(\frac{\sum x_i}{n}\right)$

With re

1 sample → 1 person

n samples → n people

6.3.1 Approximate Distribution of the Sample Mean

Let $X_1, \ldots, X_n$ be a sample from a population having mean μ and variance σ^2. The central limit theorem can be used to approximate the distribution of the sample mean

$$\bar{X} = \sum_{i=1}^{n} X_i/n$$

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Since a constant multiple of a normal random variable is also normal, it follows from the central limit theorem that $\overline{X}$ will be approximately normal when the sample size n is large. Since the sample mean has expected value μ and standard deviation $\sigma/\sqrt{n}$, it then follows that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

has approximately a standard normal distribution.

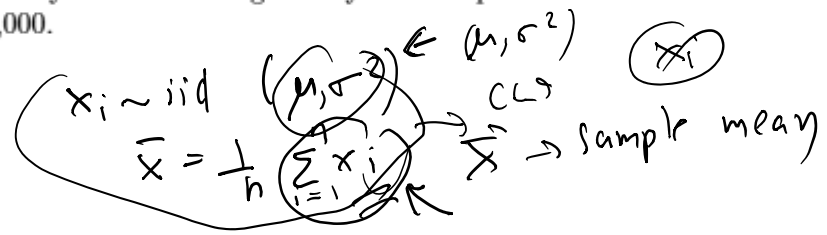
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Q9

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Q9. Average Salary of Graduates

The average salary of newly graduated chemical engineering bachelor's students is \$53,600, with a standard deviation of \$3,200. Approximate the probability that the average salary of a sample of 12 recently graduated chemical engineers exceeds \$55,000.



$E(x_i) = \mu = 53,000$

$\sigma_{x_i} = 3,200 = \sqrt{Var(x_i)}$

$\bar{X} \sim N(\mu, \sigma^2/n)$ as shown in prev qns

$\sigma_{\bar{x}} = \sigma/\sqrt{n}$

$\Rightarrow \sigma_{\bar{x}} = 3200/\sqrt{12} \approx 923.76$ //

$$P(\bar{X} > 55,000) = P\left(Z > \frac{55,000 - 53,600}{923.76}\right)$$

$$= P(Z > 1.52)$$

$$= 1 - \Phi(1.52)$$

=

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Q10

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Q10. Sample Mean of Workers' Weights

The weights of a population of workers have a mean of 167 and standard deviation 27.

(a) If a sample of 36 workers is chosen, calculate the probability that the sample mean of their weights lies between 163 and 170.

(b) Repeat part (a) for a sample size of 144.

$\mu = 167 \quad , \quad \sigma = 27$

$\bar{X} \sim N(\mu, \sigma^2/n)$

$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 27/\sqrt{36} = 4.5$

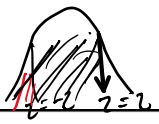
$P(163 \leq \bar{X} \leq 170) = P(\bar{X} \leq 170) - P(\bar{X} \leq 163)$

$= P\left(Z \leq \frac{170-167}{9/2}\right) - P\left(Z \leq \frac{163-167}{9/2}\right)$

$= \Phi(2/3) - \Phi(-8/9)$

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Q11
A
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A random variable X is said to be distributed as $Bernoulli(\theta)$, denoted by $X \sim Bernoulli(\theta)$, if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim Bernoulli(\theta)$, $i = 1, 2, \ldots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through Central Limit Theorem, is given by

(Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$)

(A) $\phi(2) - \phi(-2)$

(B) $\phi(1) - \phi(-1)$

(C) $\phi(3) - \phi(-3)$

(D) $\phi(90) - \phi(60)$

$P(z \leq 2) - P(z \leq -2)$
$= \phi(2) - \phi(-2)$
$\phi(2) - \phi(-2)$

CLT

iid $\rightarrow N(\mu, \frac{\sigma^2}{n}) \sim Z(0,1)$

$-2 \leq Z \leq -2$

$E(Y) = 75$

$\sigma(Y) = \sqrt{300 \times \frac{3}{4} \times \frac{1}{4}}$

$= \frac{30}{4}$

$\text{Var } X_i = p(1-p)$

$\frac{60-75}{15/2} \leq Y \leq \frac{90-75}{15/2}$

$\frac{1}{4} \cdot \frac{3}{4}$

$300 \times \frac{3}{4} \times \frac{1}{4}$

$\frac{30}{4} = 15/2$

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